

EXPERIMENT 12: TRIFILAR SUSPENSION**Aim**

Determination of acceleration due to gravity and the moment of inertia of the weight (placed on the lower disc) about the axis of rotation using trifilar suspension.

Materials and Apparatus

Two wooden discs, strings, spirit level, stopwatch, pointer, vernier caliper, divider, meter ruler, weights and two stands.

Theory

According to Figure 1, P_1 , P_2 , P_3 are three fixed points on upper disk, A, B and C are corresponding points where the strings are attached to the lower disc. When a small angular displacement is given the new positions of A, B and C are A' , B' and C' respectively.

Let O be the center of lower disc,

And
$$\widehat{AP_1A'} = \psi$$

$$\widehat{AOA'} = \phi$$

Where,

ψ : angle between the string and vertical axis

ϕ : angle swept by the radius of the disc

Considering the force equilibrium to vertical direction,

$$3f \cos \psi = Mg$$

Where,

f : tension of the string

M : mass of the lower disc

g : intensity of gravity

Since ψ is small, $\cos \psi = 1$;

$$3f = Mg \quad (1)$$

And

$$AA' = L \sin \psi$$

$$= x \sin \phi$$

Where,

x : distance from center of the disc to string

L : length of the string

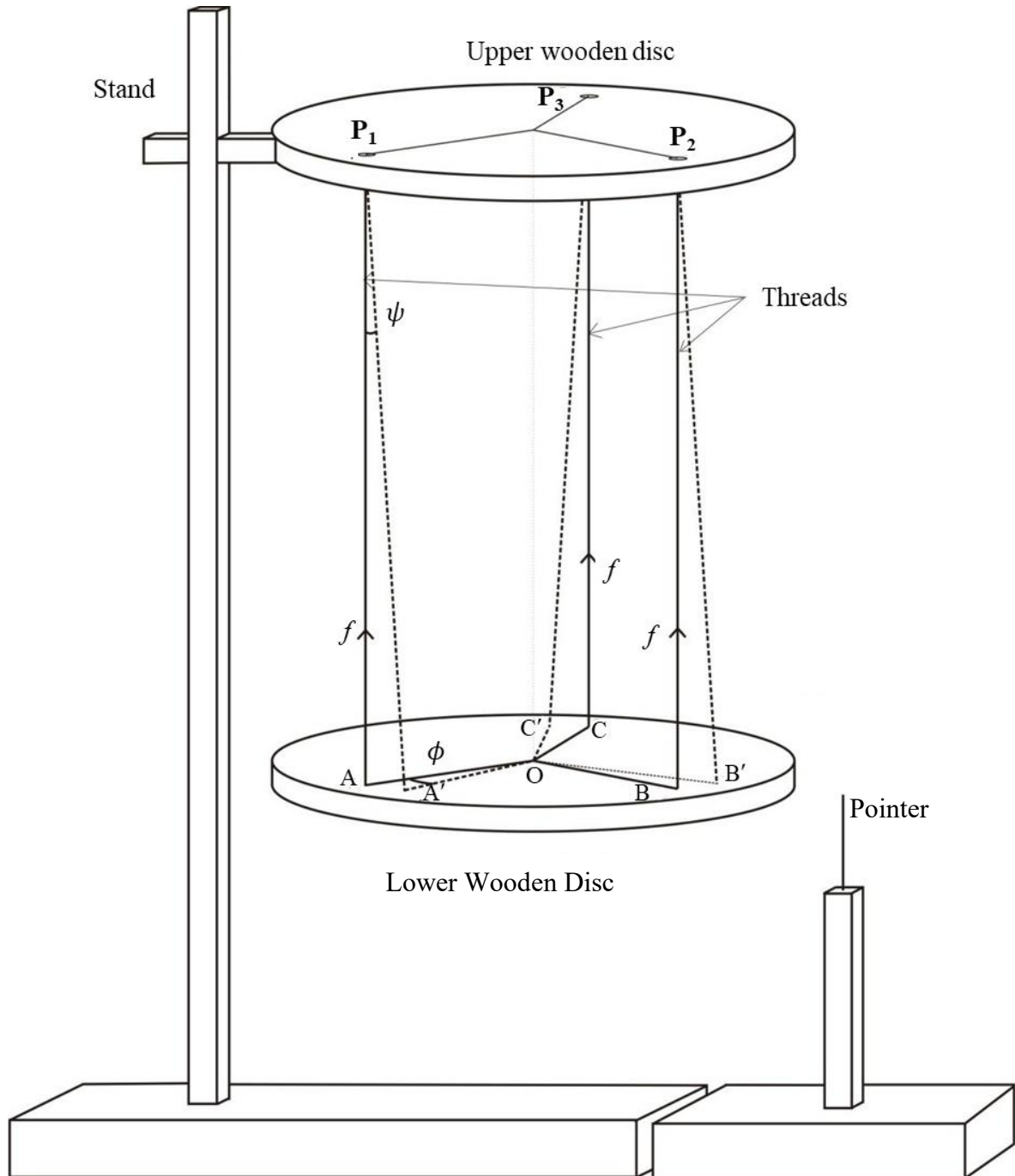


Figure 1: The setup of the experiment
 where, P_1, P_2, P_3 : Fixed points on upper disc, ψ : Angle between the string and the vertical axes, ϕ : Angle swept by the radius of the disc, f : Tension of the string, A, B, C : Corresponding points when the strings are attached, A', B', C' : New positions when a small angular displacement was given, O : Center of the lower disc

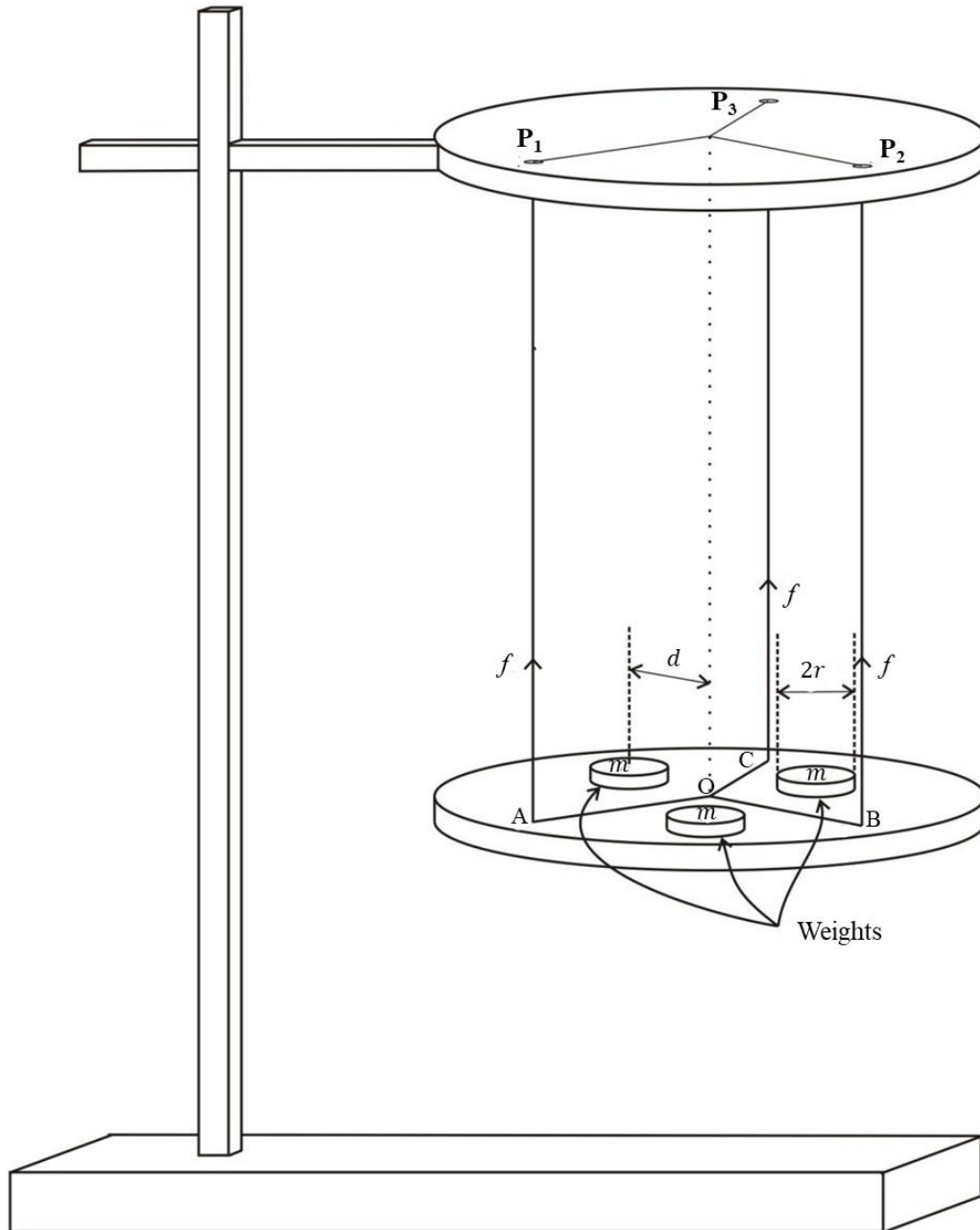


Figure 2: Placement of weights

where, d : Distance to the weight from O, $2r$: Diameter of weight, m : Mass of the weight

Since ψ and ϕ are small, $\sin \psi = \psi$ and $\sin \phi = \phi$.

Hence

$$\begin{aligned} AA' &= L\psi \\ &= x\phi \end{aligned} \quad (2)$$

The torque about the axis of rotation can be written as,

$$\tau = (3f \sin \psi)x$$

Since ψ is small,

$$\tau = 3f\psi x \quad (3)$$

From (1), (2) and (3),

$$\tau = \frac{Mgx^2}{L} \phi$$

Using $\tau = I\alpha$ for the lower disc,

$$\frac{Mgx^2}{L} \phi = -I \ddot{\phi}$$

But

$$I = \frac{1}{2}Ma^2$$

Where,

I : moment of inertia of the disc

a : radius of the disc

$$\ddot{\phi} = -\left(\frac{Mgx^2}{LI}\right) \phi$$

This is the form of the equation of harmonic motion,

Hence

$$\omega^2 = \left(\frac{Mgx^2}{LI}\right)$$

But

$$T = \frac{2\pi}{\omega}$$

Hence

$$T^2 = \left(\frac{4\pi^2 LI}{Mg}\right) \frac{1}{x^2} \quad (4)$$

Similarly if three weights are attached to the lower disc as shown in Figure 2, period of oscillation of the combined system T_c is given by,

$$T_c^2 = \left(\frac{4\pi^2 L(I + 3I_m)}{(M + 3m)g} \right) \frac{1}{x^2} \quad (5)$$

Where,

m : mass of attached weight

I_m : moment of inertia of the weight about the axis of rotation

In each case (given in equation (4) and (5)), the gradients G_1 and G_2 of the graphs T^2 vs. $\frac{1}{x^2}$ are given respectively by,

$$G_1 = \frac{4\pi^2 LI}{Mg}$$

$$G_2 = \frac{4\pi^2 L(I + 3I_m)}{(M + 3m)g}$$

Hence

$$\frac{G_2}{G_1} = \frac{(I + 3I_m)}{I} \left(\frac{M}{M + 3m} \right)$$

Thus, by knowing I , G_1 , G_2 , M , and m , I_m can be found.

Procedure

1. **Q:** What are the adjustments you should do before starting the experiment?
2. Set the distance between two discs as 1 m and keep the distance between the center of the disc and the string as 16 cm.
3. Give an angular displacement to the lower disc and measure the period of oscillation.

Q: What type of oscillation you should consider when measuring the period?

4. Repeat step 3 by decreasing the distance by 2 cm at each step.
5. Place the three weights and repeat the above procedure.
6. What are the other measurements you will need?
7. Obtain another expression for I_m using a theorem.
8. Using the above expression calculate theoretical value of I_m .

Take $M = 1.750$ kg.